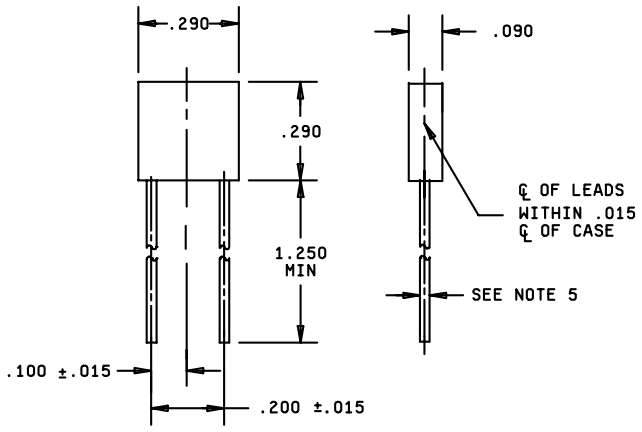
INCH-POUND
MIL-PRF-39014/2L
27 April 2004
SUPERSEDING
MIL-PRF-39014/2K
16 September 1997

PERFORMANCE SPECIFICATION SHEET

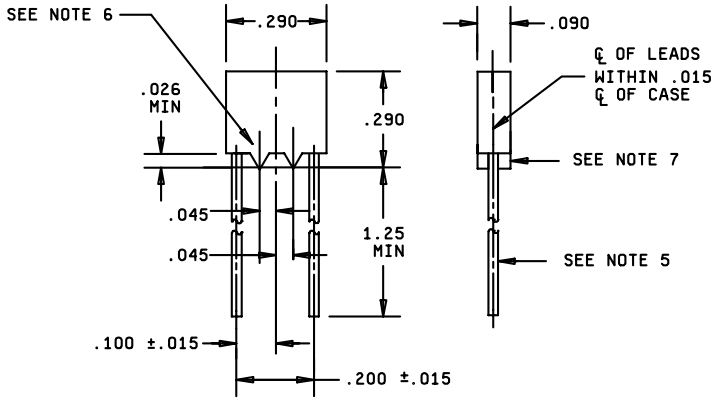
CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY, STYLE CKR06, NATO TYPE DESIGNATION NCC62

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-39014.



CONFIGURATION WITHOUT STANDOFFS



OPTIONAL CONFIGURATION WITH STANDOFFS

FIGURE 1. Dimensions and configuration.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 inch (0.25 mm).
4. Lead length may be a minimum of .625 inch (15.88 mm) for use in tape and reel automatic insertion equipment, when specified.
5. Lead diameter shall be .023 inch (0.58 mm) to .029 inch (0.74 mm).
6. Optional standoffs shall be of triangular shape so that the standoff will provide line contact with the surface upon which the capacitor is mounted.
* 7. Thickness of standoffs shall be a minimum of .075 inch (1.90 mm) to a maximum of the actual thickness of the capacitor body.
8. Insulation coating shall not extend more than .018 inch (0.46 mm) along lead wires.
9. At the option of the user, the standoff configuration may be substituted as a replacement for the nonstandoff configuration of the same style.
10. For flush mounting .078 inch (1.98 mm), printed circuit hole diameter is required to clear shoulder.

FIGURE 1. Dimensions and configurations - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Molded.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: See table I.

Radiographic inspection (failure rate level 'S' (0.001 percent/1,000 hours only)): In accordance with MIL-PRF-39014.

Dissipation factor: Not more than 2.5 percent.

Dielectric withstanding voltage:

Body insulation: Shall be able to withstand 1,300 volts between leads and case.

Insulation resistance (IR):

At 25°C: Not less than 100,000 megohms or 1,000 megohm-microfarads, whichever is less.

At 125°C: Not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.

Voltage-temperature limits: In accordance with MIL-PRF-39014.

Capacitance change with reference to +25°C	
Step A through step D of MIL-PRF-39014, voltage-temperature limit cycle table Bias = 0 volts	Step E through step G of MIL-PRF-39014, voltage-temperature limit cycle table Bias = rated voltage
±15 percent	+15 percent, -25 percent

Immersion:

IR: Not less than 50 percent of initial 25°C requirement.

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* Salt atmosphere (corrosion) (salt spray): Not applicable.

* Resistance to soldering heat: Shall be in accordance with method 210 of MIL-STD-202, test condition C.

ΔC: +15 percent, -5 percent of the initial 25°C measurement.

Solderability: The leads shall be solderable up to .020 inch (0.51 mm) from the body egress.

Workmanship: For optional standoff configuration, exposed copper or bare leads shall be permitted to a maximum of .020 inch (0.51 mm) from the body egress. For parts without standoffs, workmanship shall be in accordance with MIL-PRF-39014.

Life:

Rated conditions, 100 percent of dc rated voltage at 125°C.

0 hour through 3,000 hours:

IR: At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.

ΔC: ±20 percent from initial measured value.

Dissipation factor: Not greater than 2.5 percent.

4,000 hours through 32,000 hours:

IR: At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.

ΔC: ±20 percent from initial measured value.

Dissipation factor: Not greater than 2.5 percent.

Accelerated conditions, 200 percent of dc rated voltage at 125°C.

0 hour and 250 hours:

IR: At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.

ΔC: ±20 percent from initial measured value.

Dissipation factor: Not greater than 2.5 percent.

1,000 hours, 2,000 hours, and 4,000 hours:

IR: At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.

ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 2.5 percent.

Part or Identifying Number (PIN): Established reliability - M39014/02-(dash number from table I).
Nonestablished reliability - M39014/02C- (failure rate level M dash number from table I).

Supersession data: See table II.

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TABLE I. Dash number and electrical characteristics.

Dash number 1/ 2/				Capacitance (pF)	Capacitance tolerance (± percent)	DC rated voltage (volts)
Failure rate level (%/1,000 hours)						
1.0 (M)	0.1 (P)	0.01 (R)	0.001 (S)			
1201-	1241-	1281-	1321-	1,200	10	200
1202-	1242-	1282-	1322-	1,500	10	200
1203-	1243-	1283-	1323-	1,500	20	200
1204-	1244-	1284-	1324-	1,800	10	200
1206-	1246-	1286-	1326-	2,200	10	200
1207-	1247-	1287-	1327-	2,200	20	200
1208-	1248-	1288-	1328-	2,700	10	200
1209-	1249-	1289-	1329-	3,300	10	200
1210-	1250-	1290-	1330-	3,300	20	200
1211-	1251-	1291-	1331-	3,900	10	200
1212-	1252-	1292-	1332-	4,700	10	200
1213-	1253-	1293-	1333-	4,700	20	200
1214-	1254-	1294-	1334-	5,600	10	200
1215-	1255-	1295-	1335-	6,800	10	200
1216-	1256-	1296-	1336-	6,800	20	200
1217-	1257-	1297-	1337-	8,200	10	200
1218-	1258-	1298-	1338-	10,000	10	200
1219-	1259-	1299-	1339-	10,000	20	200
1231-	1271-	1311-	1351-	12,000	10	100
1220-	1260-	1300-	1340-	15,000	10	100
1221-	1261-	1301-	1341-	18,000	10	100
1222-	1262-	1302-	1342-	22,000	10	100
1232-	1272-	1312-	1352-	27,000	10	100
1223-	1263-	1303-	1343-	33,000	10	100
1224-	1264-	1304-	1344-	39,000	10	100
1225-	1265-	1305-	1345-	47,000	10	100
1226-	1266-	1306-	1346-	56,000	10	100
1227-	1267-	1307-	1347-	68,000	10	100
1229-	1269-	1309-	1349-	82,000	10	100
1230-	1270-	1310-	1350-	100,000	10	100
1233-	1273-	1313-	1353-	120,000	10	50
1234-	1274-	1314-	1354-	150,000	10	50
1235-	1275-	1315-	1355-	180,000	10	50
1236-	1276-	1316-	1356-	220,000	10	50
1237-	1277-	1317-	1357-	270,000	10	50
1238-	1278-	1318-	1358-	330,000	10	50
1239-	1279-	1319-	1359-	390,000	10	50
1240-	1280-	1320-	1360-	470,000	10	50
1404-	1408-	1412-	1416-	560,000	10	50
1405-	1409-	1413-	1417-	680,000	10	50
1406-	1410-	1414-	1418-	820,000	10	50
1407-	1411-	1415-	1419-	1,000,000	10	50

1/ Dash number shall include the letter "V" to indicate the optional standoff configuration (when applicable) or "-" shall be deleted for the configuration without the standoff.

2/ For nonestablished reliability (product level C) parts, the complete PIN shall include the letter “C” in place of “-“ at the beginning of the dash number and the failure rate level “M” dash number (e.g. M39014/02C1201).

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TABLE II. Substitutability data.

Dash number in MIL-C-39014/2E, /2F, /2G, /2H, /2J, MIL-PRF-39014/2K and /2L 1/					MIL-C-39014/2D					Substitutable for dash number in MIL-C-39014/2B and 2C					MIL-C-39014/2 and /2A				
L 2/	M	P	R	S	L	M	P	R	S	L	M	P	R	S	L	M	P	R	S
1361	1201	1241	1281	1321	0361	0201	0241	0281	0321	---	0201	0241	0281	0321	---	0001	0002	0003	0004
1362	1202	1242	1282	1322	0362	0202	0242	0282	0322	---	0202	0242	0282	0322	---	0005	0006	0007	0008
1363	1203	1243	1283	1323	0363	0203	0243	0283	0323	---	0203	0243	0283	0323	---	0009	0010	0011	0012
1364	1204	1244	1284	1324	0364	0204	0244	0284	0324	---	0204	0244	0284	0324	---	0013	0014	0015	0016
None	None	None	None	None	None	None	None	None	None	---	0205	0245	0285	0325	---	0017	0018	0019	0020
1366	1206	1246	1286	1326	0366	0206	0246	0286	0326	---	0206	0246	0286	0326	---	0021	0022	0023	0024
1367	1207	1247	1287	1327	0367	0207	0247	0287	0327	---	0207	0247	0287	0327	---	0025	0026	0027	0028
1368	1208	1248	1288	1328	0368	0208	0248	0288	0328	---	0208	0248	0288	0328	---	0029	0030	0031	0032
1369	1209	1249	1289	1329	0369	0209	0249	0289	0329	---	0209	0249	0289	0329	---	0033	0034	0035	0036
1370	1210	1250	1290	1330	0370	0210	0250	0290	0330	---	0210	0250	0290	0330	---	0037	0038	0039	0040
1371	1211	1251	1291	1331	0371	0211	0251	0291	0331	---	0211	0251	0291	0331	---	0041	0042	0043	0044
1372	1212	1252	1292	1332	0372	0212	0252	0292	0332	---	0212	0252	0292	0332	---	0045	0046	0047	0048
1373	1213	1253	1293	1333	0373	0213	0253	0293	0333	---	0213	0253	0293	0333	---	0049	0050	0051	0052
1374	1214	1254	1294	1334	0374	0214	0254	0294	0334	---	0214	0254	0294	0334	---	0053	0054	0055	0056
1375	1215	1255	1295	1335	0375	0215	0255	0295	0335	---	0215	0255	0295	0335	---	0057	0058	0059	0060
1376	1216	1256	1296	1336	0376	0216	0256	0296	0336	---	0216	0256	0296	0336	---	0061	0062	0063	0064
1377	1217	1257	1297	1337	0377	0217	0257	0297	0337	---	0217	0257	0297]	0337	---	0065	0066	0067	0068
1378	1218	1258	1298	1338	0378	0218	0258	0298	0338	---	0218	0258	0298	0338	---	0069	0070	0071	0072
1379	1219	1259	1299	1339	0379	0219	0259	0299	0339	---	0219\	0259	0299	0339	---	0073	0074	0075	0076
1380	1231	1271	1311	1351	0380	0231	0271	0311	0351	---	---	---	---	---	---	---	---	---	---
1381	1220	1260	1300	1340	0381	0220	0260	0300	0340	---	0220	0260	0300	0340	---	0077	0078	0079	0080
1382	1221	1261	1301	1341	0382	0221	0261	0301	0341	---	0221	0261	0301	0341	---	0081	0082	0083	0084
1383	1222	1262	1302	1342	0383	0222	0262	0302	0342	---	0222	0262	0302	0342	---	0085	0086	0087	0088
1384	1232	1272	1312	1352	0384	0232	0272	0312	0352	---	---	---	---	---	---	---	---	---	---
1385	1223	1263	1303	1343	0385	0223	0263	0303	0343	---	0223	0263	0303	0343	---	0089	0090	0091	0092
1386	1224	1264	1304	1344	0386	0224	0264	0304	0344	---	0224	0264	0304	0344	---	0093	0094	0095	0096
1387	1225	1265	1305	1345	0387	0225	0265	0305	0345	---	0225	0265	0305	0345	---	0097	0098	0099	0100
1388	1226	1266	1306	1346	0388	0226	0266	0306	0346	---	0226	0266	0306	0346	---	0101	0102	0103	0104
1389	1227	1267	1307	1347	0389	0227	0267	0307	0347	---	0227	0267	0307	0347	---	0105	0106	0107	0108
None	None	None	None	None	None	None	None	None	None	---	0228	0268	0308	0348	---	0109	0110	0111	0112
1390	1229	1269	1309	1349	0390	0229	0269	0309	0349	---	0229	02659	0309	0349	---	0113	0114	0115	0116
1391	1230	1270	1310	1350	0391	0230	0270	0310	0350	---	0230	0270	0310	0350	---	0117	0118	0119	0120
1392	1233	1273	1313	1353	0392	0233	0273	0313	0353	---	---	---	---	---	---	---	---	---	---
1393	1234	1274	1314	1354	0393	0234	0274	0314	0354	---	---	---	---	---	---	---	---	---	---
1394	1235	1275	1315	1355	0394	0235	0275	0315	0355	---	---	---	---	---	---	---	---	---	---
1395	1236	1276	1316	1356	0395	0236	0276	0316	0356	---	---	---	---	---	---	---	---	---	---
1396	1237	1277	1317	1357	0396	0237	0277	0317	0357	---	---	---	---	---	---	---	---	---	---
1397	1238	1278	1318	1358	0397	0238	0278	0318	0358	---	---	---	---	---	---	---	---	---	---
1398	1239	1279	1319	1359	0398	0239	0279	0319	0359	---	---	---	---	---	---	---	---	---	---
1399	1240	1280	1320	1360	0399	0240	0280	0320	0360	---	---	---	---	---	---	---	---	---	---
1400	1404	1408	1412	1416	0400	0404	0408	0412	0416	---	---	---	---	---	---	---	---	---	---
1401	1405	1409	1413	1417	0401	0405	0409	0413	0417	---	---	---	---	---	---	---	---	---	---
1402	1406	1410	1414	1418	0402	0406	0410	0414	0418	---	---	---	---	---	---	---	---	---	---
1403	1407	1411	1415	1419	0403	0407	0411	0415	0419	---	---	---	---	---	---	---	---	---	---

1/ Dash numbers in lower failure rate level columns are substitutable for dash numbers in all higher failure rate level columns.
2/ Revision E only.

Marking: For parts without standoffs, marking shall be in accordance with MIL-PRF-39014. For parts with standoffs, marking shall be as shown in the following example:

FRONT

M39014 2-1201 V12345

Abbreviated PIN

"V" for standoffs and manufacturer's name or CAGE code

BACK

0417A J100V 122K

Date and lot codes
"J" brand and voltage
capacitance and tolerance

Established reliability with optional standoff

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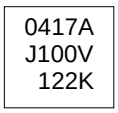
FRONT

Abbreviated PIN

"V" for standoffs and manufacturer's name or CAGE code

BACK



Date and lot codes
"J" brand and voltage
capacitance and tolerance
tolerance

Nonestablished reliability with optional standoff 1/

At the option of the manufacturer, marking may be on one side of the capacitor.

1/ NOTE: The marking requirement for nonestablished reliability (product level C) differs from the actual PIN of the nonestablished reliability part in the placement of the character "C".

Amendment notations: The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-2255)

Review activities:
* Army - AR, AT, AV, MI
Navy - MC, TD
Air Force - 19, 99

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